

Praveen Singh

Principal Control Systems Engineer
GE Vernova India · 15 years

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Gurgaon, India

PROFESSIONAL SUMMARY

Experienced OT Cybersecurity Engineer with a strong foundation in electrical engineering and 15 years of progressive responsibility. Expertise in Harmonic Analysis, Busbar Design, Commissioning & Testing. Committed to delivering safe, reliable and sustainable energy solutions.

CORE COMPETENCIES

• Harmonic Analysis	• Busbar Design	• Commissioning & Testing
• Single Line Diagram	• ETAP Software	• HMI Development
• IED Configuration	• Load Flow Analysis	• Cloud Platforms (AWS/Azure)
• DERMS	• Python Programming	• Power BI
• Data Analytics	• SQL Database	• Edge Computing
• R Programming	• Smart Metering	

PROFESSIONAL EXPERIENCE

Principal Control Systems Engineer GE Vernova India

2022 – Present

- EV Charging Network — Design and deployment of 8 EV charging stations across Gurgaon with load management integration
- Enterprise SCADA Upgrade — Migration from legacy to modern SCADA/DCS platform across 7 generation units
- Power Quality Improvement — Harmonic filter design and installation at Gurgaon reducing THD from 12% to 3.5%
- Load Forecasting Model — ML-based short-term load forecasting achieving 3% MAPE for a 15 MW utility
- Achieved forced outage rate of 12% against industry benchmark of 11%

Control Systems Engineer CESC Limited

2017 – 2022

- EV Charging Network — Design and deployment of 6 EV charging stations across Gurgaon with load management integration
- 100 MW Wind Energy Project — Turbine foundation design, inter-array cabling, substation and SCADA implementation
- Digital Twin Implementation — Real-time digital twin of 17 MW thermal plant for predictive maintenance and optimization
- Led team of 20+ engineers across 4 project locations

Digital Solutions Engineer WBSEDCL

2013 – 2017

- Digital Twin Implementation — Real-time digital twin of 9 MW thermal plant for predictive maintenance and optimization
- OT Cybersecurity Hardening — ICS security assessment and remediation across 10 substations per IEC 62351 standards
- Commissioned 17 MW capacity ahead of schedule by 9 weeks

Digital Solutions Engineer MSEDCL

2012 – 2013

- Distribution Network Strengthening — R-APDRP project covering 4 feeders with automated fault detection and restoration
- Substation Automation — IEC 61850 based protection and control implementation at 8 EHV substations

- 50 MW Utility Solar PV Plant at Gurgaon — EPC project from design to commissioning including SCADA integration and grid synchronization
- 400kV GIS Substation at Gurgaon — Design, procurement and commissioning of 20×315 MVA auto-transformers with numerical protection

EDUCATION

B.Tech (Electrical & Instrumentation)

2009

NIT Warangal

CERTIFICATIONS & TRAINING

- ✓ Solar PV Design Certification (MSME)
- ✓ Certified Energy Auditor (BEE, Government of India)
- ✓ Siemens S7 PLC Certification

References available upon request · Last updated: March 2026